How We Compute FTP Charges, Credits, and Realised Margin

Once an account has a `CHOSEN_WAL` (see `docs/behavioural_wal.md`, `docs/prepayment_wal.md`, `docs/persistence_structural_split_wal.md`) and a rate curve to look it up against (`docs/base_curve_construction.md`, `docs/liquidity_premium_construction.md`), here's how we turn that into an actual daily charge, credit, and realised margin figure per account — and how we roll those up into our consolidation-level totals.

Rate assignment

We set `FTP_RATE` for an account to the curve looked up at its `CHOSEN_WAL` — `BASE_RATE + LP`, both interpolated from our fitted curves at that exact tenor point. Separately, we resolve `OBS_RATE_FINAL` — the account's real, observed interest rate — through its own cascade of tiers (an explicit rate source, an interest-accrual-derived rate, or a documented fallback of last resort), consulting a manual override sheet first wherever we have a specific correction on file.

Account-level pricing

We compute four figures per account (or per tranche, for structural- split accounts — see `docs/persistence_structural_split_wal.md`):

```
PRICING_GAP                 = OBS_RATE − FTP_RATE          (only where a real, non
BASE_CHARGE_DAILY_ACCRUAL   = TRANCHE_OUT_BAL × BASE_RATE × (1 / 365)
LP_CHARGE_DAILY_ACCRUAL     = TRANCHE_OUT_BAL × LP         × (1 / 365)
FTP_CHARGE_DAILY_ACCRUAL    = TRANCHE_OUT_BAL × FTP_RATE   × (1 / 365)
DAILY_PRODUCT_MARGIN_VALUE = TRANCHE_ABS_OUT_BAL × PRICING_GAP × (1 / 365)
```

`FTP_CHARGE_DAILY_ACCRUAL` is, by construction, the sum of the base and LP charges — the FTP rate itself is base + LP, so charging at the full FTP rate is equivalent to charging the base and liquidity components separately and adding them. We keep both the combined figure and the two components so our daily-consolidation layer can report where a period's total FTP charge actually came from — pure term cost vs. liquidity cost — not just the total.

We deliberately leave `PRICING_GAP` and the margin value that depends on it blank (not zero) wherever there's no meaningful observed rate to compare against. An account with a genuinely missing observed rate still gets FTP-charged — charging is driven purely by `CHOSEN_WAL`, independent of whether an observed rate exists — but its *realised margin* is undefined without one, not zero.

Rolling up to consolidation

At each consolidation level (daily → weekly → monthly → segment), we produce three kinds of figures from the account-level numbers above:

Weighted averages, recomputed from the period's own totals rather than averaged across sub-periods directly:

```
WTD_AVG_OBS_RATE            = Σ(OBS_RATE × balance) / Σ(balance)
WTD_AVG_FTP_RATE            = Σ(FTP_RATE × balance) / Σ(balance)
WTD_[period]_PRICING_GAP    = WTD_AVG_OBS_RATE − WTD_AVG_FTP_RATE
```

We recompute the pricing gap at a rolled-up level from that level's own weighted rates, rather than averaging the daily pricing gaps directly — the two aren't equivalent once balances change day to day within the period, and the recomputed version is the one that stays consistent with the level's own reported weighted rates.

Totals, straightforward sums across every account/tranche in scope: `TOTAL_BASE_CHARGE_DAILY`, `TOTAL_LP_CHARGE_DAILY`, `TOTAL_FTP_CHARGE_DAILY`, `TOTAL_MARGIN_DAILY`.

A transfer-direction call, comparing the total FTP charge against a configured neutrality tolerance, so we can report whether a branch or segment is a net payer or net receiver of internal funds transfer pricing for the period, or genuinely neutral within tolerance.

Why this should (mostly) net to zero

We designed FTP as a double-entry, zero-sum mechanism: Treasury sits as the mirror counterparty to every business-side charge or credit. A perfectly matched book — assets and liabilities of identical size and tenor — nets to exactly zero. A real, non-zero net position across the whole bank is what we'd expect, not a defect: it's the quantified cost (or benefit) of genuine maturity transformation, the bank funding long-dated assets with shorter-dated liabilities or vice versa. That's a materially different thing from a net position caused by a **data gap** — a product that simply never reached FTP pricing at all, which shows up as a real imbalance for a reason that has nothing to do with genuine tenor mismatch. Telling the two apart is a reconciliation exercise, not something our pricing formulas resolve on their own — it's worth building dedicated diagnostics for (product-mapping-gap tracking, consistencyof-gap-direction checks across branches and segments) rather than assuming every non-zero net position tells the same story.

Where we hit real limits

- **Realised margin needs a genuine observed rate.** Any systematic gap in our observed-rate capture for a product understates how much of the book has a knowable realised margin, independent of whether FTP charging itself is working correctly.
- **Zero-sum is a design target, not a guarantee.** As above — genuine tenor mismatch, data-mapping gaps, and classification issues can all produce a non-zero net position, and each needs a different fix. Treating any non-zero figure as automatically “the mismatch cost” (or automatically “a bug”) is a common but real risk we’ve had to guard against explicitly in any reporting built on top of these totals.

See also `docs/behavioural_wal.md`, `docs/base_curve_construction.md`, `docs/liquidity_premium_construction.md`.